Gaussian Mixture Model Based Pseudo-label Learning for Semisupervised Hyperspectral Image Classification

Wei Li, Zhuomin Zhou, Wei Fu, Ting Lu, *Member, IEEE*, and Xudong Kang, *Senior Member, IEEE*

***Abstract*—Semisupervised learning can alleviate the heavy reliance on labeled samples for hyperspectral image (HSI) classification while suffering from the critical problem of the accurate identification of reliable pseudo-labeled data. The problem is mainly caused by the fact that the model may predict an incorrect label with high confidence and vice versa, especially in the case of limited labeled samples. To solve this problem, a new Gaussian mixture model based pseudo-label learning method (GMM-PLL) is proposed for HSI classification (HSIC). First, we innovatively model the reliability of classification results using a Gaussian mixture model modeling (GMM-M) module. Benefiting from this, reliable pseudo-labeled samples can be adaptively selected based on the probabilistic model, rather than a rigid reliance on prediction confidence. Then, considering that a limited number of labeled samples cannot sufficiently train the classification model, we construct a class-balanced contrastive learning (CBCL) mechanism to fully exploit unlabeled information for training. As a consequence, pseudo-label sample generation and classification model training are performed alternately, which produces more reliable pseudo-labels and better models. Finally, satisfactory classification performance can be achieved based on the well-trained model. Experimental results demonstrate the effectiveness and superiority of the proposed method. The code is available at https://github.com/zzm-hnu/GMM-PLL**

***Index Terms*—Classification, contrastive loss, Gaussian mixture model (GMM), hyperspectral image (HSI), semisupervised learning.**

I. Introduction

HYPERSPECTRAL images (HSIs), characterized by high spectral resolution and broad wavelength coverage, provide rich spectral information that enables accurate classification of land cover in complex observation scenarios. Consequently, HSI classification is widely applied in fields such as land use [1], environmental monitoring [2], and national defense [3]. However, due to the high spectral dimensions, complex spatial structures, and difficulty in acquiring ground-truth data, effective feature learning to improve classification accuracy remains a significant challenge.

In recent years, there has been a transition in HSI classification (HSIC) from traditional approaches [4], [5], [6], [7], [8] to deep learning-based methods. Chen et al. [9] first used CNN to extract regularized deep features in HSIC. Cheng et al. [10] proposed a dual-branch deep stochastic adaptive Fourier decomposition network to extract deep features for classification. Cheng et al. [11] further proposed a deep high-order tensor sparse representation network, which can obtain better classification results in the case of small training samples. Furthermore, many scholars have introduced attention mechanisms into their research to encourage the network to learn more valuable information. For instance, Yu et al. [12] proposed the center-specific perception transformer network, which is the first attempt to perform class-guided attention and filter interference factors for HSIC feature representation. He et al. [13] utilized a CNN to extract spatial features of HSIs and an improved transformer to capture sequential spectral relationships. However, the heavy computational burden of self-attention mechanisms limits the potential of transformers in real-time applications. On the contrary, the recently proposed Mamba architecture provides a more efficient solution for HSIC by leveraging selective state spaces to capture global context with linear complexity. For instance, Zu et al. [14] propose text-guided contrastive Mamba (Mamba-TGC), a novel text-guided contrastive learning framework for HSIC. Zhao et al. [15] propose SliMamba, a lightweight convolution-Mamba architecture for HSIC. Meng et al. [16] introduced a spatial-frequency joint learning Mamba (SFMamba) framework for HSIC. Furthermore, extensive studies have investigated classification models with limited labeled samples. These methods can be broadly categorized into three types: transfer learning-based methods, generative adversarial learning-based methods, and few-shot learning-based methods. Transfer learning-based methods focus on cross-domain classification, leveraging a source domain with sufficient labeled samples to transfer learned knowledge to the target domain, thereby achieving accurate classification [17], [18], [19], [20]. Generative adversarial learning-based methods aim to generate highly realistic samples to augment the training set for effective classification [21], [22], [23]. Few-shot learning-based methods explore the differences in feature distributions among various classes and extract more discriminative feature representations from limited labeled samples [24], [25], [26].

However, the above deep learning methods belong to the supervised learning paradigm, in which performance is

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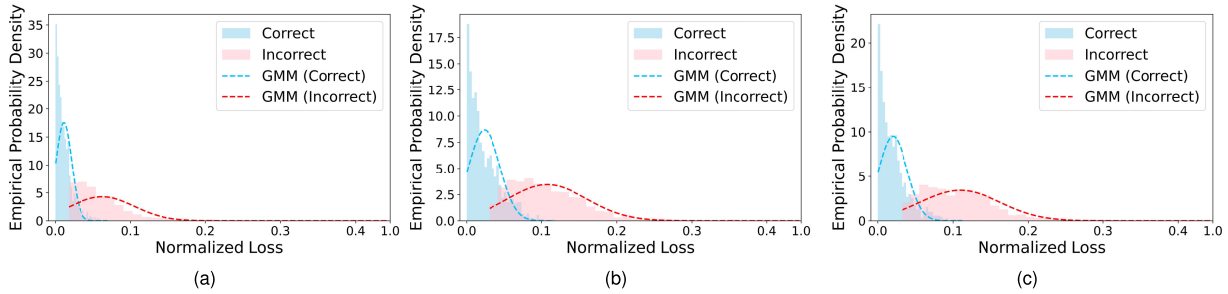


Fig. 1. Distribution of loss values for (a) PaviaU, (b) Houston 2013, and (c) Indian_Pines.

negatively affected when there are insufficient training samples. To eliminate the dependence on labeled samples, some deep learning methods began to explore semisupervised learning strategies in the HSIC task. In particular, semisupervised learning methods seek to improve performance by using both labeled and unlabeled samples, which can be broadly classified into three categories, including graph-based methods, reconstruction-based methods, and pseudo-label-based methods. Graph-based methods [27], [28] achieve classification objectives by constructing adjacency matrices to capture the intrinsic structure of both labeled and unlabeled samples. For example, Zhang et al. [29] proposed dual sparse representation graph-based collaborative propagation for HSIC. They adopted sparse representation to construct spectral and spatial graphs on spectral and spatial dimensions. Then, label propagation is performed on the two graphs iteratively. Reconstruction-based methods [30] generally employ autoencoders to learn feature representations of unlabeled samples in an unsupervised manner while simultaneously fine-tuning network parameters using labeled samples in a supervised manner. For example, Jia et al. [31] integrated an autoencoder module and a Siamese network to, respectively, investigate information in a large amount of unlabeled data and rectify it with a limited labeled sample set. Pseudo-label-based methods [32], [33], [34], [35] train a classifier on labeled samples to classify unlabeled ones, then select high-confidence predictions as pseudo-labels, which are combined with the labeled set to iteratively retrain the classifier. However, graph-based methods struggle to fully represent the detailed spatial structure of HSIs via graph structures and suffer from high computational complexity. Meanwhile, reconstruction-based models rely on specific constraints to reconstruct the original HSIs, which is difficult to implement in practice. On the contrary, pseudo-label-based methods have been widely studied due to their simple principle and promising performance.

Based on the above analysis, we attempted to design a novel pseudo-label-based semisupervised learning method for HSIC. In this article, there are two main issues we would like to address, i.e., the potential generation of erroneous pseudo-labels and the underutilization of information from unlabeled samples. Existing pseudo-label-based methods generally select reliable pseudo-labels according to the prediction probabilities of models, and then train the network with labeled samples and samples with reliable pseudo-labels, while samples with unreliable pseudo-labels are discarded. However, filtering reliable pseudo-labels via prediction probabilities suffers from difficulties in threshold adjustment and the risk of introducing erroneous information; furthermore, useful information contained in unlabeled samples is not fully exploited. When we observe the loss value distribution during training, it reveals that correctly classified samples usually exhibit low loss values, whereas misclassified samples show high loss values. As shown in Fig. 1, statistical analysis of loss distributions in three different datasets during training reveals that this pattern is consistent. Moreover, the loss distribution curves for correctly and incorrectly classified samples tend to approximate two distinguishable Gaussian curves.

To address the negative impacts of erroneous pseudo-label generation and the underutilization of unlabeled samples, we propose a Gaussian mixture model based pseudo-label learning method (GMM-PLL) method for semisupervised HSIC. We target performance improvement from two perspectives: reliable sample set partitioning and informative mining from unreliable samples. First, we construct two branches: a weak augmentation branch and a strong augmentation branch. The weak augmentation branch applies weak augmentation to both labeled and unlabeled samples, and generates pseudo-labels by leveraging the potential distribution consistency between labeled and unlabeled spectral pixels. The strong augmentation branch applies strong augmentation to unlabeled samples and utilizes the intrinsic consistency of the same sample to design a consistency constraint [36] for pseudo-labels in model training. Then, we design a GMM modeling (GMM-M) module to select reliable pseudo-labeled samples. Unlike previous pseudo-label selection strategies that typically choose samples with high prediction probabilities [37], [38], [39], we employ a Gaussian mixture model (GMM) to model the loss distribution and determine the reliability of pseudo-labels via posterior probabilities, thereby selecting reliable pseudo-labels for unlabeled samples. Finally, considering that unreliable pseudo-labeled samples still contain substantial useful information, we design a class-balanced contrastive learning (CBCL) mechanism. This mechanism pulls features of the same sample under different augmentation conditions closer while pushing features of samples from different classes apart, enabling the feature extractor to learn more discriminative features. In summary, the main contributions of our method are as follows.

1) It is the first time to model the reliability of pseudo-labels using the GMM. The idea is based on the observation that loss values of misclassified samples approximately follow a Gaussian distribution due to spectral mixing, while a similar pattern is observed for correctly classified samples. These two Gaussian distributions form a GMM, which can be used to derive the posterior probability, thereby adaptively selecting reliable pseudo-labels of unlabeled samples.
2) We designed a CBCL mechanism to leverage the information of unlabeled samples for training. Features of the same sample should be similar under different augmentation conditions, while features of samples from

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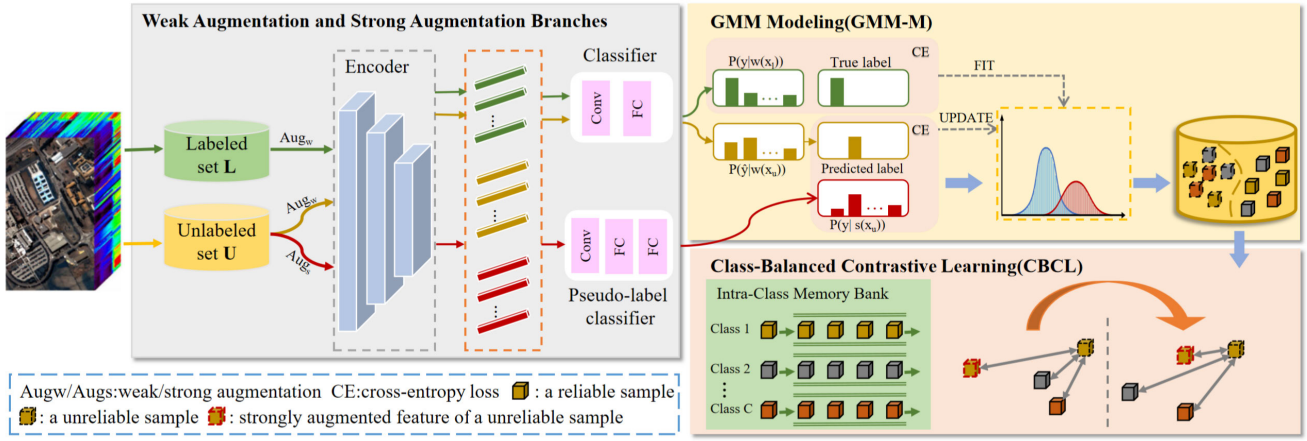


Fig. 2. Flowchart of the proposed method GMM-PLL.

different classes should be distinguishable. Therefore, we constructed an intraclass memory bank, storing the features of samples from each class. By constraining the features of unreliable samples, we ensure that features remain similar under different augmentation conditions and are separated from those of samples belonging to other categories. This mechanism enables the model to learn more discriminative features, thereby facilitating the effective training of the classification model.

3) A union semisupervised classification framework for HSIC is developed by constructing weak augmentation and strong augmentation branches. The weak augmentation branch exploits the latent distribution consistency between labeled spectral pixels and unlabeled spectral pixels to produce pseudo-labels. The strong augmentation branch leverages the intrinsic consistency of the same sample to design a consistency constraint of pseudo-labels for model training, and provides self-supervised information for contrastive learning.

The remainder of this article is organized as follows. Section II presents the details of the proposed method. Section III gives experimental results and analysis of several publicly available datasets. Finally, we summarize our work in Section IV.

II. Methodology

A. Overview

The GMM-PLL method aims to improve the correctness of generated pseudo-labels and efficiently exploit unlabeled data in semisupervised learning scenarios, thereby enhancing classification performance. In particular, we first construct a dual-branch framework consisting of weak and strong augmentation. The first branch applies weak augmentation $w(\cdot)$ to both the labeled sample $x_l \in L$ and the unlabeled sample $x_u \in U$; features are extracted by the encoder and passed through a classifier to output predictions $P(y|w(x_l))$ and $P(\hat{y}|w(x_u))$. The second branch applies strong augmentation $s(\cdot)$ to the unlabeled sample $x_u \in U$, where the encoder extracts features that are then processed by a pseudo-label classification head to output predictions $P(y|s(x_u))$. This dual-branch structure imposes consistency constraints on the generated pseudo-labels and provides self-supervised information for contrastive learning. On this basis, we design a GMM-M for selecting reliable pseudo-labeled samples. Leveraging the loss distribution patterns of labeled data, we employ a two-component GMM [40] to dynamically filter reliable pseudo-labeled samples for training, while iteratively updating the loss distribution using the loss values from this selected set. Meanwhile, we design a class-balanced contrastive loss module for model training. For unreliable pseudo-labeled samples, we utilize reliable features stored in the intraclass memory bank to construct a contrastive loss, which enhances feature discriminability and further improves the performance of the classification model.

Finally, we integrate the supervised loss $\mathcal{L}_l$ from labeled samples, the unsupervised loss $\mathcal{L}_u$ from reliable pseudo-labeled samples, and the contrastive loss $\mathcal{L}_{ctr}$ from unreliable pseudo-labeled samples to drive continuous learning and progressively boost classification performance. The flowchart of the proposed method GMM-PLL is shown in Fig. 2. Detailed descriptions are given below.

B. GMM Modeling Module for Reliable Pseudo-label Sample Selection

Based on observations of the loss distribution during training, as shown in Fig. 1, we conducted a statistical analysis of loss values across three different datasets. We found that the loss distribution reflects the reliability of classification results to a certain extent: correctly classified samples tend to exhibit lower loss values, whereas misclassified samples correspond to higher ones. Notably, the loss distribution curves for correct and incorrect samples approximate two distinguishable Gaussian curves. Therefore, we propose an effective GMM-M module to partition the reliable pseudo-labeled samples.

The GMM [41] is a probabilistic generative model that models complex data distributions through a combination of multiple Gaussian distributions, thereby revealing latent cluster structures within the data. GMMs have been widely applied in scenarios such as semisupervised learning, clustering, and probability density estimation [42], [43], [44]. The probability density function of a GMM is defined as a linear weighted sum of multiple unimodal Gaussian components. In this article, we fit a two-component GMM using the supervised loss $\mathcal{L}_l$ of all labeled samples to model the loss distribution patterns of low-loss (i.e., reliable pseudo-labeled samples) and high-loss (i.e., unreliable pseudo-labeled samples) categories, respectively. Let π denote the weight of the corresponding component, satisfying $\sum_{k=1}^{2} \pi_k = 1$. Then, the probability

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density function of this two-component GMM is defined as follows:

$$P(x \mid \theta) = \sum_{k=1}^{2} \pi_k \cdot \mathcal{N}\left(x \mid \mu_k, \sigma_k^2\right) \tag{1}$$

where θ denotes the overall parameter set of the GMM and $\mathcal{N}(x \mid \mu, \sigma^2)$ denotes the probability density function of the Gaussian component.

Based on the probability density function $P(x|\theta)$, we proceed to select reliable pseudo-labeled samples. For an unlabeled sample $x_u \in U$ with a predicted probability distribution $P(\hat{y}|w(x_u))$, we designate the class with the highest probability as the predicted label, which is defined as

$$\hat{y} = \arg\max \boldsymbol{P}(\hat{y} \mid \omega(x_u)) \tag{2}$$

where $\hat{y}$ is the predicted label. Subsequently, we calculate the cross-entropy loss L_u' between $\hat{y}$ and the prediction $P(y|s(x_u))$ of the strongly augmented unlabeled sample $x_u \in U$, which is defined as

$$\mathcal{L}_u' = F(\hat{y}, \boldsymbol{P}(\hat{y} \mid s(x_u))) \tag{3}$$

where $F(\cdot)$ denotes the cross-entropy loss function. We input $\mathcal{L}_u$ into the GMM fit in the current round to obtain the posterior probability that the sample belongs to the low-loss (i.e., reliable pseudo-labeled samples) component, which is defined as

$$\gamma_t(\mathcal{L}_u, t) = \frac{\pi_t \cdot \mathcal{N}\left(\mathcal{L}_u \mid \mu_t, \sigma_t^2\right)}{\sum_{j=1}^{2} \pi_j \cdot \mathcal{N}\left(\mathcal{L}_u \mid \mu_j, \sigma_j^2\right)} \tag{4}$$

where t denotes the Gaussian component of "low loss (i.e., reliable pseudo-labeled samples)." This probability value represents the likelihood that a pseudo-labeled sample is reliable. Since the quality of pseudo-labeled samples may gradually decline as the number of samples increases with each training iteration, we have established a dynamically adjusted threshold that increases linearly to assess sample reliability. If a sample's GMM posterior probability exceeds this threshold, it is deemed a reliable pseudo-labeled sample. The threshold is specifically defined as

$$\alpha = \alpha_l + (\alpha_h - \alpha_l)/n_e \tag{5}$$

where α_l is the initial low threshold, α_h is the final high threshold, and n_e is the total number of epochs.

During training, we periodically update the GMM to adapt to changes in the distribution of loss values. In the early stages of training, we use only the cross-entropy loss values of labeled samples under the current model as fitting data; in later stages, we incorporate the cross-entropy loss of unlabeled samples into the fitting data.

C. CBCL Mechanism for Classification Model Training

Contrastive learning, which learns discriminative features by constructing effective positive and negative sample pairs, has been increasingly applied in the field of data classification [45], [46], [47], [48], [49]. In particular, contrastive learning causes the features of positive pairs to be close to each other while pushing the features of negative pairs apart. The contrastive loss is defined as follows:

$$\mathcal{L}_{InfoNCE} = \frac{1}{n} \sum_{i=1}^{n} -\log \frac{e^{\mathrm{sim}\left(d_i, d_i^+\right)/\tau}}{e^{\mathrm{sim}\left(d_i, d_i^+\right)} + \sum_{j=1}^{m} e^{\mathrm{sim}\left(d_i, d_j^-\right)/\tau}} \tag{6}$$

where d_i is the feature representation of a sample, d_i^+ is the feature representation of a positive sample, m is the number of negative samples, d_i^- is the feature representation of a negative sample, $sim(\cdot)$ denotes the cosine similarity function, and τ is the temperature hyperparameter.

To fully exploit the unlabeled data, we do not directly discard unreliable pseudo-labeled samples. Instead, we continue to extract information from them through a CBCL mechanism, thereby further training the classification model.

In this module, we construct an intraclass memory bank containing feature queues for all classes, which are used to store the features of samples from each category. For class c, its class feature queue is defined as follows:

$$\mathbf{M}_c = \begin{bmatrix} f_{c,1}^\top \\ f_{c,2}^\top \\ \vdots \\ f_{c,K}^\top \end{bmatrix} \in \mathbb{R}^{K \times D} \tag{7}$$

where K is the capacity of the feature queue, D is the feature dimension, and $f_{c,i}^\top$ is the transpose of the features of the ith reliable sample in class c.

In each training round, features from labeled samples are added first. When new reliable pseudo-labeled samples are generated, their features are also added in real time to the queue of the corresponding category. If a queue reaches its maximum capacity, the oldest features are automatically removed. This length constraint not only maintains the balance of sample numbers across categories but also ensures that the memory bank retains high-quality, up-to-date features, providing a stable and reliable supervision reference for the feature learning of unreliable samples. Finally, the intraclass memory bank tensor, composed of feature queues from all classes, is denoted as

$$\mathbf{T} = \begin{bmatrix} M_1 \\ M_2 \\ \vdots \\ M_C \end{bmatrix} \in \mathbb{R}^{C \times K \times D} \tag{8}$$

where C denotes the total number of categories in the dataset.

For an unreliable sample x_{ur}, its weakly augmented feature is defined as follows:

$$d_i = \mathrm{Encoder}(w(x_{ur})) \tag{9}$$

where $\omega(\cdot)$ denotes a weak augmentation operation and $Encoder(\cdot)$ denotes a feature extraction operation. The strongly augmented feature for the unreliable sample x_{ur} is defined as follows:

$$d_i^+ = \mathrm{Encoder}(s(x_{ur})) \tag{10}$$

where $s(\cdot)$ denotes a strong augmentation operation.

The d_i^+ is incorporated into the contrastive learning module as a positive sample for d_i, while negative samples are obtained from the intraclass memory bank. Let $\hat{c}$ denote the predicted label for an unreliable sample; then, the negative feature tensor D^- consists of features from all feature queues excluding the class $\hat{c}$, where a specific negative sample $d_i^- \in D^-$. In particular, D^- is defined as

$$\mathcal{D}^- = \bigcup_{\substack{c=1 \\ c \neq \hat{c}}}^{C} \mathcal{M}_c \tag{11}$$

where C denotes the total number of categories in the dataset.

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In the proposed method, a threshold β is set. If the posterior probability that the GMM classifies a sample as a reliable pseudo-labeled sample is less than β, the sample is treated as an unreliable sample and subjected to contrastive learning. The contrastive loss is defined as follows:

$$\mathcal{L}_{ctr} = \frac{1}{N_d}\sum_{i=1}^{N_d} -\log \frac{e^{\mathrm{sim}\left(d_i, d_i^{+}\right)/\tau}}{e^{\mathrm{sim}\left(d_i, d_i^{+}\right)} + \sum_{j=1}^{m} e^{\mathrm{sim}\left(d_i, d_j^{-}\right)/\tau}} \quad \text{s.t.,}\ \gamma_{it}\left(\mathcal{L}_{ui}, t\right) < \beta \tag{12}$$

where N_d is the number of unreliable samples.

This CBCL mechanism enhances the consistency of feature extraction for unreliable samples and improves interclass discrimination by pulling positive pairs closer and pushing negative pairs apart. In particular, the module utilizes the features of reliable samples in the memory bank as anchors to help unreliable samples clearly define their boundaries with other classes, thereby reducing the classification error rate and further training the classification model.

D. Overall Loss

The loss function of GMM-PLL comprises three components: the supervised loss $\mathcal{L}_l$ from labeled samples, the unsupervised loss $\mathcal{L}_u$ from reliable pseudo-labeled samples, and the contrastive loss $\mathcal{L}_{ctr}$ from unreliable pseudo-labeled samples. The mathematical formula is expressed as follows:

$$\mathcal{L} = \mathcal{L}_l + \mathcal{L}_u + \sigma \cdot \mathcal{L}_{ctr} \tag{13}$$

where σ is the preset weighting coefficient. $\mathcal{L}_l$ and $\mathcal{L}_u$ are expressed, respectively, as

$$\mathcal{L}_l = \frac{1}{N_l}\sum_{i=1}^{N_l} F\left(y_i, \boldsymbol{P}\left(y_i \mid \omega\left(x_{li}\right)\right)\right) \tag{14}$$

$$\mathcal{L}_u = \frac{1}{N_u}\sum_{i=1}^{N_u} F\left(\hat{y}_i, \boldsymbol{P}\left(\hat{y}_i \mid s\left(x_{ui}\right)\right)\right) \quad \text{s.t.,}\ \gamma_{it}\left(\mathcal{L}_{ui}, t\right) > \alpha \tag{15}$$

where $F(\cdot)$ is the cross-entropy function. N_l and N_u are the total number of labeled samples and reliable pseudo-labeled samples. α is the reliability threshold.

III. Experiments

In this section, we describe the conducted experiments to demonstrate the effectiveness of the proposed GMM-PLL for HSIC. First, we briefly describe the dataset used in the experiments. Next, we present the experimental setup and parameter discussion, followed by an ablation study. Finally, we perform comparison experiments of our method with some state-of-the-art methods and discuss the impact of varying training sample sizes on the experimental results.

A. Dataset Description

Three widely used HSI datasets are utilized in the experiments, including the University of Pavia (PaviaU), Houston 2013, and Indian_Pines. We will give a short introduction to the datasets below.

1) *PaviaU:* This dataset was acquired by the ROSIS-03 sensor of the University of Pavia, Italy. It covers a spatial area of 610×340 pixels with a spatial resolution of 1.3 m per pixel and a spectral range from 0.43 to 0.86 μm. It originally comprised 115 contiguous spectral bands. To ensure data quality and eliminate interference, 12 noisy or redundant bands were removed, leaving 103 valid bands for subsequent research and algorithm validation. The dataset includes nine distinct land cover categories with 42 776 labeled samples, excluding the background category.
2) *Houston 2013:* This dataset was acquired by the ITRES CASI-1500 sensor, covering the University of Houston campus and its surrounding areas. The spatial coverage spans 349×1905 pixels with a spatial resolution of 2.5 m per pixel. Spectral coverage ranges from 0.364 to 1.046 μm, comprising 144 contiguous spectral bands. The dataset includes 15 land cover classes with 15 029 labeled samples, excluding background categories.
3) *Indian_Pines:* This dataset was acquired by the AVIRIS sensor over an experimental site in northwestern Indiana. It covers a spatial area of 145×145 pixels with a spatial resolution of 20 m per pixel and spectral coverage from 0.4 to 2.5 μm. It originally comprised 220 bands. After excluding 20 bands affected by water absorption and noise, the remaining 200 bands were retained. The dataset includes 16 distinct classes with 10 249 labeled samples, excluding background classes.

B. Experiment Setup

In our experiments, the proposed GMM-PLL method was implemented on a single NVIDIA GeForce RTX 3090 GPU using the PyTorch framework. For each class, ten labeled samples were randomly selected as the training set, and the remaining samples were used as the test set. The details of the training and test samples of the PaviaU, Houston 2013, and Indian_Pines datasets are listed in Table I. To quantitatively and qualitatively evaluate the classification performance, four metrics were adopted: overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and class accuracy (CA). Each experiment was repeated ten times, and the final results were obtained by averaging the outcomes of the ten runs.

The same parameter settings were applied to all datasets. The model was trained using the Adam optimizer with a learning rate of 0.001. The warm-up stage lasted for 20 epochs, followed by 200 training epochs. The GMM was updated every 5 epochs. The loss values of unlabeled samples were included in the fitting process starting from epoch 20. The memory queue length for each class was set to 100. The hyperparameters to be set include the temperature hyperparameter τ in the contrastive loss, the contrastive loss weighting coefficient σ, and the unreliability threshold β. Based on the parameter discussion, they are set to 0.10, 0.001, and 0.5, respectively. It should be noted that our experiments employ a consistent parameter setting for all datasets. Two data augmentation strategies were used. Weak augmentation included random horizontal and vertical flipping. Strong augmentation introduced random noise on the basis of weak augmentation.

C. Parameter Discussion

For the proposed GMM-PLL method, three hyperparameters are analyzed in this section. These include the temperature hyperparameter τ in the contrastive loss, the weighting coefficient σ, and the unreliability threshold β. The parameters τ

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TABLE I

NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS IN PAVIAU, HOUSTON 2013, AND INDIAN_PINES DATASETS

PaviaU				Houston 2013				Indian_Pines			
No.	Class Name	Train	Test	No.	Class Name	Train	Test	No.	Class Name	Train	Test
1	Asphalt	10	6621	1	Healthy grass	10	1241	1	Alfalfa	10	36
2	Meadows	10	18639	2	Stressed grass	10	1244	2	Corn-notill	10	1418
3	Gravel	10	2089	3	Synthetic grass	10	687	3	Corn-mintill	10	820
4	Trees	10	3054	4	Trees	10	1234	4	Corn	10	227
5	Metal Sheets	10	1335	5	Soil	10	1232	5	Grass-pasture	10	473
6	Bare Soil	10	5019	6	Water	10	315	6	Grass-trees	10	720
7	Bitumen	10	1320	7	Residential	10	1258	7	Grass-pasture-mowed	10	18
8	Bricks	10	3672	8	Commercial	10	1234	8	Hay-windrowed	10	468
9	Shadows	10	937	9	Road	10	1242	9	Oats	10	10
	Total	90	42686	10	Highway	10	1217	10	Soybean-nottill	10	962
				11	Railway	10	1225	11	Soybean-mintill	10	2445
				12	Parking Lot 1	10	1223	12	Soybean-clean	10	583
				13	Parking Lot 2	10	459	13	Wheat	10	195
				14	Tennis Court	10	418	14	Woods	10	1255
				15	Running Track	10	650	15	Buildings-Grass-Trees-Drives	10	376
					Total	150	14879	16	Stone-Steel-Towers	10	83
									Total	160	10089

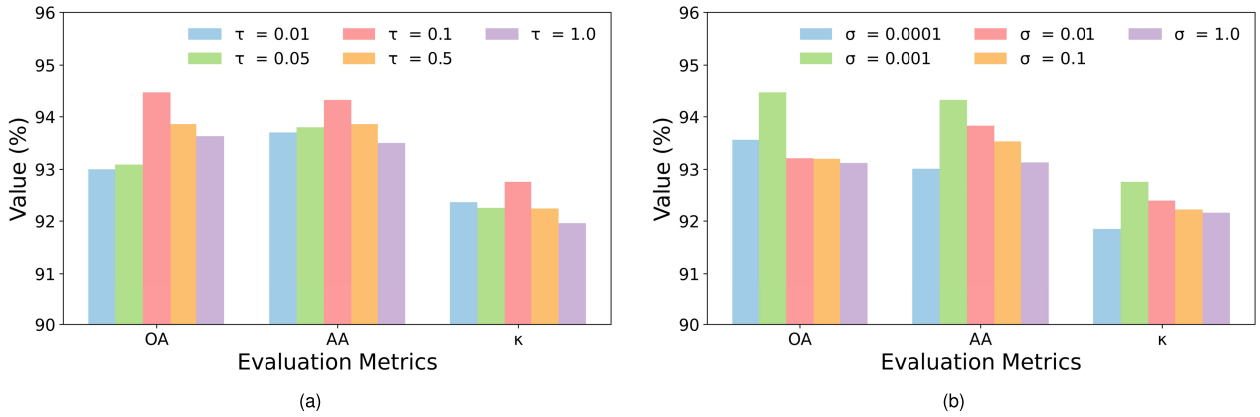


Fig. 3. Effect of different parameters on experimental results on the PaviaU dataset. (a) τ. (b) σ.

and σ influence the effectiveness of contrastive learning, while β affects the number and quality of unreliable pseudo-labeled samples. These factors ultimately influence the classification performance of the model.

1) *Temperature Hyperparameter τ and Weighting Coefficient σ of the CBCL Mechanism:* τ is used to adjust the degree to which the contrastive loss module distinguishes between positive and negative samples. The smaller the value of τ, the stronger the model's ability to distinguish between positive and negative samples, but this increases the risk of overfitting; conversely, the larger the value of τ, the more difficult it is for the model to learn discriminative features. Experiments were conducted with different τ values, including 0.01, 0.05, 0.10, 0.50, and 1.00. Fig. 3(a) depicts the impact of different values of parameter τ on classification performance; the model achieves the highest OA, AA, and κ when $\tau = 0.10$, indicating that the classification performance is optimal at this point. The parameter σ controls the contribution of contrastive loss in the overall loss function. A larger σ makes the model focus more on learning discriminative features. At the same time, it weakens the primary classification objective. Therefore, an appropriate σ value is necessary. Fig. 3(b) illustrates the influence of different values of σ on classification performance. When $\sigma = 0.001$, OA, AA, and κ achieve the best results.
2) *Unreliability Threshold β:* The parameter β represents the selection criterion for unreliable pseudo-labeled samples. A larger β increases the likelihood of a sample being correctly classified, which may lead to the erroneous inclusion of samples into the unreliable set. On the contrary, a smaller β results in fewer selected unreliable samples, thereby providing limited discriminative information. Based on this analysis, β should not be set too large or too small. Fig. 4 presents the influence of β on classification performance, which shows that the best performance is obtained when $\beta = 0.5$.

D. Ablation Studies

The GMM-M module and CBCL Mechanism are key components of the proposed method. Ablation experiments were designed to verify their effectiveness. The experimental results on three different datasets are presented in Table II, and the best results in the table have been indicated in bold.

GMM-M: In this experiment, we follow a conventional semisupervised strategy. Reliable pseudo-labeled samples are selected based on the model prediction probability. The threshold is set to 0.95. The loss distribution is not modeled in this setting. The method is named w/o (i.e., without) GMM-M.

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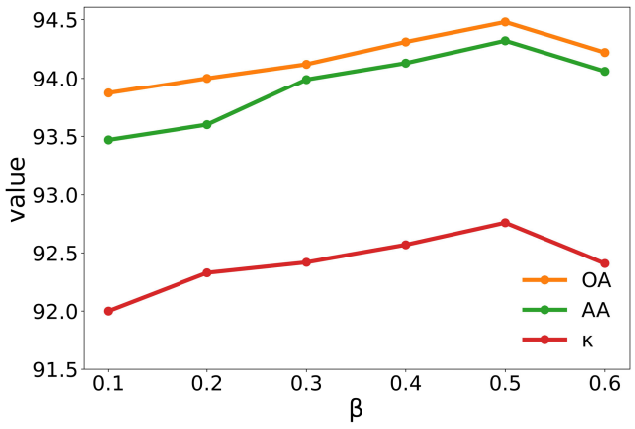


Fig. 4. Effect of different values of parameter β on experimental results on the PaviaU dataset.

TABLE II
COMPARISON OF QUANTITATIVE METRICS(%) OF ABLATION EXPERIMENTS ON PAVIAU, INDIAN PINES, AND HOUSTON 2013 DATASETS

Dataset	Metrics	w/o GMM-M	w/o CBCL	GMM-PLL
paviaU	OA	93.06±2.88	93.54±2.48	**94.48±1.70**
	AA	93.30±1.91	93.58±1.63	**94.32±1.54**
	κ	90.93±3.66	91.55±3.17	**92.76±2.19**
Houston 2013	OA	89.36±1.41	89.58±1.36	**90.34±1.63**
	AA	90.70±1.24	90.86±1.18	**91.55±1.44**
	κ	88.50±1.53	88.74±1.47	**89.56±1.76**
Indian_Pines	OA	85.37±2.57	85.21±2.11	**86.28±2.22**
	AA	91.08±1.31	91.11±1.07	**91.36±1.54**
	κ	83.43±2.87	83.25±2.36	**84.42±2.50**

TABLE III
COMPARATIVE CLASSIFICATION RESULTS FOR EACH CLASS OF THE "W/O GMM-M" AND THE PROPOSED GMM-PLL METHODS

PaviaU			Houston 2013		
No.	w/o GMM-M	GMM-PLL	No.	w/o GMM-M	GMM-PLL
1	84.10±8.28	**87.07±8.03**	1	87.58±4.77	**88.40±5.43**
2	94.46±4.17	**95.98±2.79**	2	89.26±5.51	**89.48±6.20**
3	96.39±1.58	96.16±2.71	3	97.02±2.29	**97.76±1.54**
4	87.33±3.61	**87.80±2.51**	4	90.81±3.51	**91.62±3.45**
5	99.27±1.20	**99.75±0.50**	5	99.38±1.00	**99.58±0.81**
6	99.01±1.97	**99.63±0.54**	6	91.56±8.09	**92.16±7.98**
7	99.44±0.62	**99.59±0.65**	7	86.79±4.81	**89.94±4.46**
8	94.22±8.56	**96.03±3.46**	8	66.95±7.65	**68.95±7.48**
9	85.45±8.62	**86.85±8.47**	9	86.04±4.06	**87.88±3.53**
			10	98.01±2.47	**98.58±2.57**
			11	89.41±4.87	**90.58±5.09**
			12	84.84±11.97	**84.96±12.01**
			13	96.73±2.40	**96.88±2.51**
			14	100.00±0.00	99.98±0.07
			15	96.12±3.58	**96.58±3.56**

The results are presented in the third column of Table II. As can be seen, GMM-PLL achieves better values on three comprehensive evaluation metrics of OA, AA, and κ. In particular, the OA of GMM-PLL is 1.42%, 0.98%, and 0.91% higher than the "w/o GMM-M" on the three datasets, verifying the effectiveness of GMM-M. Furthermore, to illustrate the effect of the GMM-M module in improving classification performance, Table III presents comparative classification results for each class of the "w/o GMM-M" and the proposed GMM-PLL methods on the PaviaU and Houston 2013 datasets. In particular, on the PaviaU dataset, eight out of nine categories show improvement. On the Houston 2013 dataset, 14 out of 15 categories achieve better results.

CBCL: In this setting, the CBCL module is removed. Unreliable pseudo-labeled samples are directly discarded. The final training loss only contains $\mathcal{L}_l$ and $\mathcal{L}_u$. The method is named w/o (i.e., without) CBCL. The results indicate that all three evaluation metrics of "w/o CBCL" decrease compared with the proposed method. On the contrary, the proposed method utilizes unreliable pseudo-labeled samples through contrastive loss. This strategy enriches the training information and further improves the discriminative ability of the learned features.

E. Performance Comparison

To evaluate the effectiveness of GMM-PLL, we compare it with other advanced HSIC methods, including DMSGer [50], RSEN [51], DSCA-Net [52], EMS-GCN [53], CMLPL [54], and CACL [55]. The above methods are briefly introduced as follows.

DMSGer: This method constructs a dynamic multiscale graph convolutional network. It first builds a compact graph structure at the region level through hyperpixel segmentation and metric learning. Then, a multiscale graph convolution structure is applied. This design fully exploits spatial–spectral information at different scales. All hyperparameters follow the optimal settings reported in the original paper.

RSEN: This method proposes a robust self-integrated network. The framework contains two components: a base network and an integrated network. The model is trained using supervised loss from labeled data and consistency constraints from unlabeled data. This strategy improves classification performance. All hyperparameters follow the optimal settings reported in the original paper.

DSCA-Net: This method develops a dual-stream category-adaptive network. It uses superpixel-level label propagation to achieve balanced category sampling. It also introduces a dynamic category-adaptive threshold strategy. This mechanism alleviates classification bias between easy and difficult samples. All hyperparameters follow the optimal settings reported in the original paper.

EMS-GCN: This method maps pixel features into the superpixel feature space. It dynamically optimizes superpixel boundaries during network training. The model captures local spectral–spatial information within superpixels. It also models long-range relationships between different superpixels. Finally, the enhanced superpixel features are mapped back to the pixel space to obtain fine pixel-level classification results. All hyperparameters follow the optimal settings reported in the original paper.

CMLPL: This method reduces the negative influence of pseudo-label noise on feature learning. It improves feature discrimination through a mutual learning mechanism and a pseudo-label smoothing strategy. In addition, a dynamic threshold strategy is proposed. This strategy adaptively controls the number of unlabeled samples introduced during training. It helps reduce the interference caused by unreliable predictions in the early training stage. All hyperparameters follow the optimal settings reported in the original paper.

CACL: This method is designed for open-set semisupervised HSIC. It introduces a position-aware transformer (PAT). The PAT extracts discriminative spectral–spatial features by

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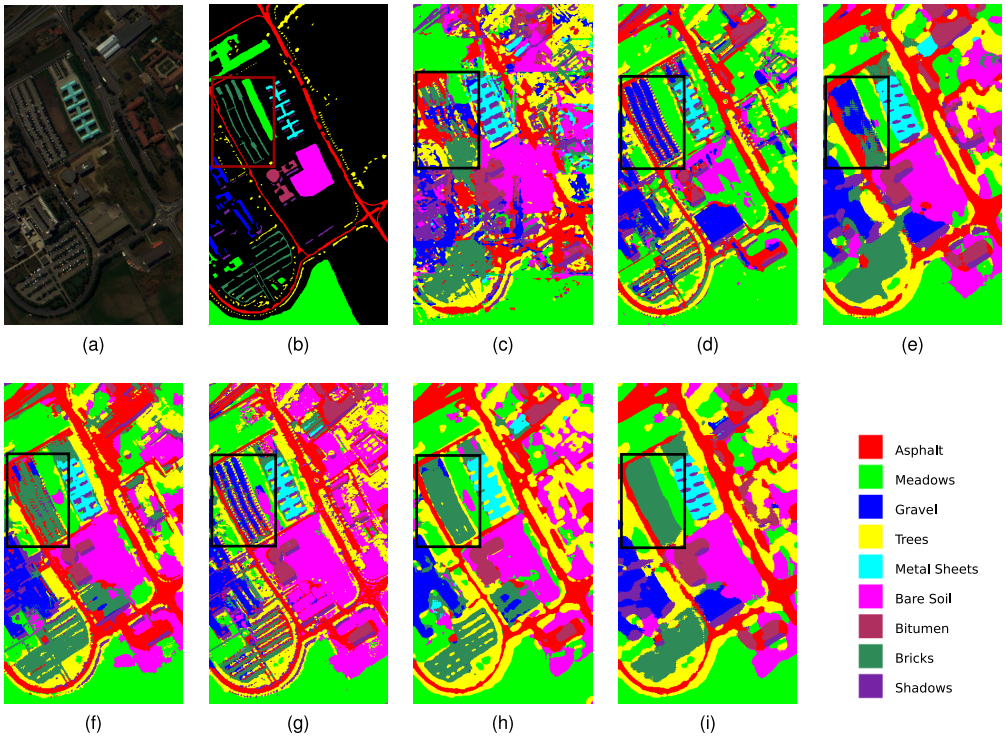


Fig. 5. Classification maps for the PaviaU dataset. (a) False color image. (b) Ground truth. (c) DMSGer. (d) RSEN. (e) DSCA-Net. (f) EMS-GCN. (g) CMLPL. (h) CACL. (i) GMM-PLL.

TABLE IV
CLASSIFICATION RESULTS(%) FOR PAVIAU DATASET WITH TEN LABELED SAMPLES PER CLASS

No.	DMSGer	RSEN	DSCA-Net	EMS-GCN	CMLPL	CACL	GMM-PLL
1	58.94±8.66	72.65±14.03	86.60±8.70	85.39±5.12	86.26±15.34	**94.24±4.06**	87.07±8.03
2	82.33±5.86	88.89±9.78	**96.03±4.04**	89.21±6.55	89.85±3.11	92.26±6.75	95.98±2.79
3	85.08±10.14	79.48±20.03	**96.51±2.34**	96.26±1.05	87.20±8.39	89.81±8.04	96.16±2.71
4	76.47±7.85	**98.41±0.68**	87.00±4.27	94.78±2.68	94.16±3.49	92.67±4.83	87.80±2.51
5	99.89±0.15	87.82±9.25	99.51±0.70	99.90±0.13	**99.99±0.02**	99.22±1.01	99.75±0.50
6	90.56±6.51	65.88±8.38	**96.94±6.25**	99.60±0.42	91.93±6.90	95.47±4.17	99.63±0.54
7	94.95±5.07	78.86±14.63	99.57±0.71	**99.86±0.19**	98.75±2.45	99.10±1.39	99.59±0.65
8	79.82±8.24	77.84±16.83	94.07±7.52	95.40±1.94	84.05±11.60	94.32±3.24	**96.03±3.46**
9	93.13±4.01	94.58±9.83	85.35±7.27	85.57±7.97	**99.66±0.39**	64.00±14.11	86.85±8.47
OA	80.82±3.24	82.12±5.19	93.87±2.31	93.78±3.08	90.18±2.64	92.83±3.35	**94.48±1.70**
AA	84.58±2.50	82.29±1.94	93.51±1.16	94.00±1.52	92.41±1.34	91.23±2.44	**94.32±1.54**
κ	75.32±3.91	77.23±6.16	91.95±2.95	91.97±3.88	87.26±3.28	90.74±4.20	**92.76±2.19**

modeling symmetric position-aware spatial prior relationships between the center pixel and its neighboring pixels. The method also utilizes prototype learning to fully exploit information from known categories. For a fair comparison with the semisupervised method proposed in this article, we configure CACL under a closed-set semisupervised setting. In this case, all unlabeled samples are assumed to belong to known categories.

Table IV presents OA, AA, κ, and classification accuracies for all categories on the PaviaU dataset. For each line, the bold font indicates the best result. In general, the proposed GMM-PLL achieves the best performance in terms of the three objective evaluation metrics. In particular, Fig. 5 presents the classification maps produced by different methods on the PaviaU dataset. It can be observed that all methods roughly recover the main feature distribution of the scene. However, other methods exhibit more noisy points and class misclassification in complex regions. In addition, the category boundaries are not sufficiently clear. On the contrary, the classification map produced by the proposed GMM-PLL is spatially smoother. It also shows stronger internal consistency within the same class region. Moreover, the boundaries between different categories are more accurately delineated. For example, in the "Bricks" area located in the upper left region, DMSGer misclassifies part of the region as "Gravel" and "Asphalt." RSEN and CMLPL misclassify most of the area as "Gravel." DSCA-Net, EMS-GCN, and CACL partially misclassify this region as "Gravel." In comparison, the proposed GMM-PLL correctly classifies this area and produces clear category boundaries. This result demonstrates the superiority of GMM-PLL in feature discrimination and classification performance.

Table V presents the objective metrics for each method for the Houston 2013 dataset. For each line, bold text indicates

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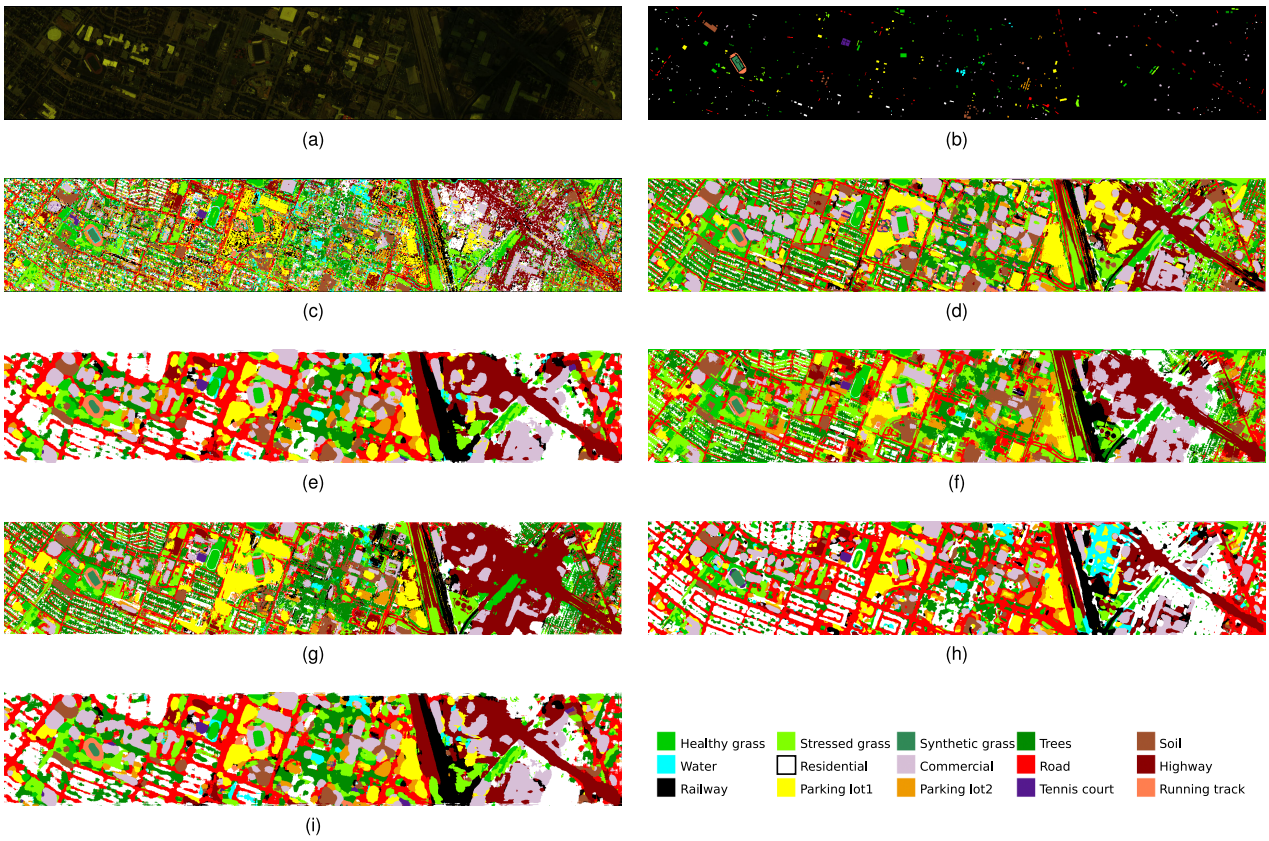


Fig. 6. Classification maps for the Houston 2013 dataset. (a) False color image. (b) Ground truth. (c) DMSGer. (d) RSEN. (e) DSCA-Net. (f) EMS-GCN. (g) CMLPL. (h) CACL. (i) GMM-PLL.

TABLE V
CLASSIFICATION RESULTS(%) FOR HOUSTON 2013 DATASET WITH TEN LABELED SAMPLES PER CLASS

No.	DMSGer	RSEN	DSCA-Net	EMS-GCN	CMLPL	CACL	GMM-PLL
1	87.07±4.04	**90.27±5.09**	87.48±6.35	89.70±4.39	88.44±5.25	87.93±5.56	88.40±5.43
2	63.10±3.28	**96.11±5.14**	89.76±5.04	89.28±7.16	87.87±6.46	90.97±9.52	89.48±6.20
3	99.33±0.95	99.61±0.94	96.80±1.84	**100.00±0.00**	**100.00±0.00**	96.97±4.81	97.76±1.54
4	54.93±9.63	**96.35±3.34**	90.92±3.82	95.29±4.72	93.91±4.15	93.20±5.31	91.62±3.45
5	98.74±1.09	**99.74±0.46**	99.49±0.65	99.65±0.60	99.13±1.28	94.29±6.32	99.58±0.81
6	**95.74±5.89**	86.52±7.95	91.33±7.85	95.14±6.42	92.76±7.85	91.11±7.90	92.16±7.98
7	67.97±9.53	89.42±3.56	88.94±3.81	87.13±5.95	83.99±5.51	85.95±14.04	**89.94±4.46**
8	47.86±5.43	**76.12±6.00**	67.61±7.24	68.98±8.49	59.64±5.70	63.39±6.28	68.95±7.48
9	66.85±5.94	83.55±6.41	86.08±4.22	86.89±6.86	82.05±7.94	85.71±8.90	**87.88±3.53**
10	83.86±10.94	83.68±11.23	97.46±4.65	94.30±5.53	76.92±14.01	89.46±6.51	**98.58±2.57**
11	80.69±9.92	77.02±7.00	90.08±6.21	85.48±7.03	71.59±11.15	83.16±8.99	**90.58±5.09**
12	71.13±9.30	84.04±13.05	**85.96±11.62**	78.65±12.65	79.70±11.83	78.14±13.08	84.96±12.01
13	86.00±4.52	63.28±6.02	95.96±3.27	84.90±6.75	79.48±6.81	90.92±8.29	**96.88±2.51**
14	**100.00±0.00**	86.71±16.75	**100.00±0.00**	99.83±0.43	**100.00±0.00**	96.34±4.87	99.98±0.07
15	99.01±1.43	99.90±0.14	95.61±2.94	**99.98±0.05**	**99.98±0.05**	77.48±29.51	96.58±3.56
OA	76.48±1.74	87.95±1.28	89.65±1.38	89.08±1.74	84.56±2.38	86.04±1.48	**90.34±1.63**
AA	80.15±1.65	87.41±1.80	90.61±1.65	90.35±1.64	86.36±2.36	87.00±1.62	**91.55±1.44**
κ	74.79±1.91	86.96±1.38	88.82±1.36	88.19±1.88	83.32±2.57	84.95±1.59	**89.56±1.76**

the best performance. The proposed GMM-PLL achieves the highest accuracy in 5 of the 15 categories. It also obtains the best performance in OA, AA, and κ. Fig. 6 shows the classification maps of different methods on the Houston 2013 dataset. The comparison methods exhibit more point noise in the classification maps. Moreover, the spatial distribution within the same class region appears discontinuous. The classification results are therefore less smooth. On the contrary, the proposed GMM-PLL produces better classification maps. This improvement is achieved by filtering more reliable pseudo-labeled samples through Gaussian distributions. At the same time, the method effectively utilizes unreliable pseudo-labeled data.

Table VI presents the evaluation metrics on the Indian_Pines dataset, including OA, AA, κ, and the accuracy of each category. The best result in each row is shown in bold. From the numerical results, the proposed GMM-PLL achieves the highest OA (86.28%), AA (91.36%), and κ (84.42%).

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TABLE VI
CLASSIFICATION RESULTS(%) FOR INDIAN PINES DATASET WITH TEN LABELED SAMPLES PER CLASS

No.	DMSGer	RSEN	DSCA-Net	EMS-GCN	CMLPL	CACL	GMM-PLL
1	**100.00±0.00**	97.89±2.89	99.72±0.89	94.44±4.97	**100.00±0.00**	96.94±4.02	99.72±0.83
2	56.31±4.01	39.07±12.29	69.36±9.49	69.72±11.36	59.72±9.67	74.53±7.95	**76.94±6.94**
3	73.26±6.94	60.47±11.95	76.08±7.27	77.29±10.10	75.35±13.19	**88.39±7.71**	77.49±7.19
4	93.75±5.78	71.49±16.73	**98.22±2.54**	84.63±17.08	83.26±10.98	82.98±11.77	97.53±3.97
5	79.61±4.60	67.60±15.78	87.17±7.42	88.18±5.55	84.02±6.16	**90.70±5.49**	86.13±7.51
6	81.70±6.63	97.18±1.43	95.05±1.85	88.26±7.76	**98.36±1.22**	59.79±11.14	94.67±2.62
7	**100.00±0.00**	99.75±0.50	**100.00±0.00**	91.11±5.09	**100.00±0.00**	82.39±9.13	**100.00±0.00**
8	98.48±2.58	98.06±2.50	99.58±0.58	99.72±0.25	99.72±0.45	95.42±4.27	**99.96±0.09**
9	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	78.00±25.61	**100.00±0.00**	99.44±1.67	**100.00±0.00**
10	72.90±5.78	68.18±11.42	81.08±5.18	82.27±9.07	72.93±6.99	**94.87±6.60**	79.90±8.30
11	55.80±4.54	73.40±11.13	87.00±4.68	83.91±7.45	69.09±8.60	**98.00±6.00**	86.46±5.29
12	72.44±8.48	57.42±11.36	76.10±10.26	78.87±9.27	72.33±13.59	**81.99±5.40**	78.25±10.33
13	98.97±1.70	99.49±0.00	98.72±2.00	99.18±0.95	**99.74±0.26**	81.35±7.66	99.23±1.44
14	87.59±7.40	75.66±12.24	**95.61±7.17**	94.89±8.72	87.16±7.48	82.18±8.33	92.69±8.19
15	92.07±6.20	73.15±13.54	95.43±8.96	97.42±7.03	93.80±15.85	**99.03±1.60**	95.69±6.88
16	**99.40±1.11**	91.62±8.53	97.78±2.42	95.42±3.92	98.55±2.14	88.39±7.71	97.11±4.18
OA	73.49±2.27	69.17±2.87	85.89±2.80	84.49±3.34	77.56±2.51	79.55±4.32	**86.28±2.22**
AA	85.14±1.27	79.81±1.83	91.28±1.54	87.71±2.05	87.13±1.39	87.20±2.75	**91.36±1.54**
κ	70.76±2.43	65.62±2.76	83.22±3.08	82.40±3.75	74.67±2.75	76.33±4.94	**84.42±2.50**

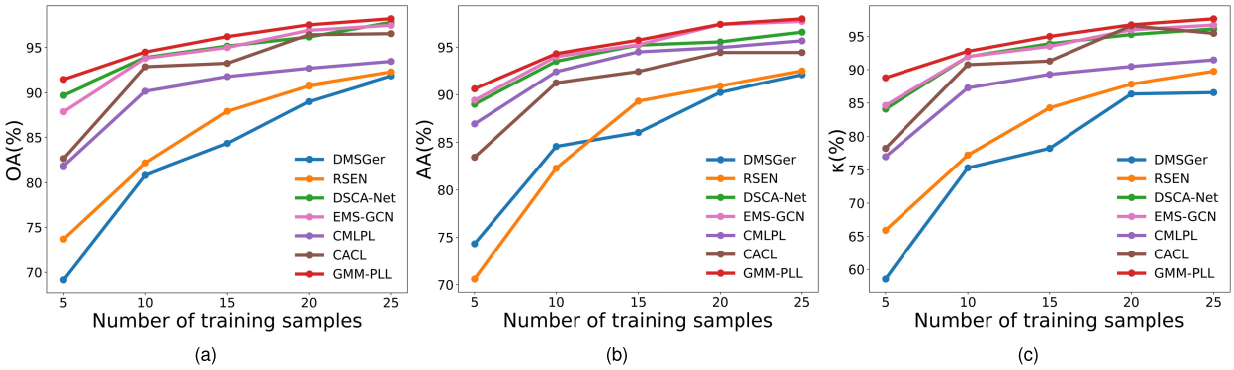


Fig. 7. Overall accuracies with respect to different numbers of training samples for the PaviaU dataset. (a) OA. (b) AA. (c) κ.

TABLE VII
MODEL COMPLEXITY OF GMM-PLL

	Parameters(M)	FLOPs(G)	Inference Time(ms)
PaviaU	82.1	183.8	84
Houston 2013	85.7	216.3	267
Indian_Pines	80.3	148.6	37

In comparison, RSEN obtains the lowest OA (69.17%) and the lowest κ (65.62%). DMSGer obtains the lowest AA (79.81%). In addition, the proposed method achieves the highest accuracy in 4 of the 16 categories. These results verify the effectiveness of the proposed method.

To comprehensively evaluate the computational cost of GMM-PLL, we present the model complexity metrics of the proposed method on three different datasets, as presented in Table VII. The results show that the number of model parameters remains stable between 80M and 86M, indicating that the scale of the model's underlying architecture is relatively fixed. It is noticed that the inference time of the proposed method varies across different datasets; e.g., the inference time is 267 ms on the Houston 2013 dataset and 37 ms on the Indian Pines dataset. Nevertheless, the overall inference time remains below 1 s, demonstrating that the model complexity of the proposed method is acceptable.

F. Effect of Training Sample Numbers

In addition, we evaluate the classification performance of the above methods with different numbers of training samples. Fig. 7 presents the evaluation metrics, including OA, AA, and κ, for the proposed method and the comparison methods. In each category, the number of training samples varies from 5 to 25, increasing gradually in steps of 5. In general, the OA, AA, and κ of all methods improve as the number of labeled samples increases. This is because more training samples provide more reliable supervised information. Such information helps improve the prediction accuracy. It can also be observed that GMM-PLL consistently achieves superior performance on all metrics. This advantage becomes more significant when fewer training samples are used. Overall, these experimental results clearly demonstrate the effectiveness of the GMM-PLL method. They also verify its better robustness and generalization ability in the case of limited training samples.

IV. CONCLUSION

This article proposes a GMM-PLL. By generating more reliable pseudo-labels and further leveraging large amounts of unlabeled data, it effectively improves classification performance. To address two challenges in semisupervised classification methods based on pseudo-labels, i.e., the unreliability of pseudo-label generation and the underutilization of

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unlabeled samples, we designed a GMM-M module to generate more reliable pseudo-labels. In addition, we developed a CBCL mechanism to fully exploit the information in unlabeled samples, thereby further training the classification model. Both theoretical analysis and experimental results demonstrate the effectiveness of GMM-PLL over some state-of-the-art methods.

Although the proposed GMM-PLL uses a GMM to simulate the distribution of loss values, the underlying distribution requires further investigation. In future work, we may consider models that better align with real-world distribution patterns to select reliable pseudo-labels, thereby improving the overall classification accuracy of the model.

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